

GAMESPHERE

Online Gaming Website

Play. Compete. Conquer.

Product Requirements Document

Project ID	WD-001
Client	GameSphere Digital Solutions Pvt. Ltd.
Location	Bengaluru, Karnataka, India
Platform	Responsive Web Application
Document Version	v1.0
Date	June 2026

1. Project Overview

1.1 Problem Statement

Casual gamers lack a single, unified browser-based destination where they can discover and play multiple HTML5 games, track personal progress, and compete with others - all without installing any software. Existing gaming portals are either cluttered with ads, require subscriptions, or do not offer community engagement features such as leaderboards and achievement systems.

1.2 Product Vision

GameSphere is a centralized online gaming portal that makes browser-based casual gaming social, trackable, and rewarding. It gives players a single account, a curated game catalog, automatic achievement recognition, and transparent leaderboards - delivered through a fast, responsive web interface accessible on any device.

1.3 Target Users

Primary audience: Casual gamers aged 16 to 35 based in India who prefer browser-based entertainment on mobile and desktop devices. Secondary audience: The GameSphere platform administrator who manages game content and user accounts through a dedicated dashboard.

2. Key Features

Feature	Description
User Authentication	Email and password registration, login, logout, and JWT-based session management.
Game Catalog	Browsable and searchable catalog of at least 6 embedded HTML5 games with title, thumbnail, description, and genre tag.
Game Search and Filter	Live search by game title and genre-based filtering via dropdown or tab selector.
Score Tracking	Automatic recording and display of per-user high score per game; updates when a new session score exceeds the stored value.
Achievement Badges	Five badges awarded automatically based on in-platform actions: First Win, High Roller, Game Explorer, Favorite Collector, Leaderboard Legend.
Favorites System	Users can mark games as favorites and view all saved favorites on a dedicated Favorites page.
Leaderboards	Per-game leaderboard (top 10 by high score) and global leaderboard (top 10 by cumulative score) with rank, username, avatar, and score.
User Profile	Profile page showing username, avatar, total games played, total score, and earned badges. Users can upload an avatar and update their display name.
Admin Dashboard	Protected admin interface to add or remove games, view registered users, and deactivate accounts.

3. User Flow

3.1 New User Journey

1. User lands on the homepage and sees the game catalog and a registration prompt.
2. User registers with email and password, is logged in automatically, and arrives at the catalog.
3. User browses or searches for a game, clicks its card, and plays it on the game detail page.
4. The game session ends; the score is recorded and the leaderboard updates if the score qualifies.
5. Earned badges appear on the user's profile page automatically based on activity triggers.
6. User can add games to favorites, view the global leaderboard, and update their profile at any time.

3.2 Admin Journey

- Admin navigates to the admin login URL and authenticates with admin credentials.
- Admin views the user list, adds new games to the catalog, removes outdated games, or deactivates user accounts.
- Changes to the game catalog are immediately reflected on the public-facing portal.

4. Functional Requirements

4.1 Authentication

- Users must register with a valid email address and a password of at least 8 characters.
- Passwords must be stored as hashed values; plain-text storage is not permitted.
- JWT tokens must be used for session management; tokens must expire after 24 hours.
- Deactivated accounts must be blocked at login and may not access any authenticated routes.

4.2 Game Catalog

- The catalog must display at least 6 HTML5 games at launch.
- Each game card must show: title, thumbnail image (minimum 300 x 200 px), short description, and genre tag.
- Live search must filter game cards in real time as the user types in the search field.
- Genre filter must show only games matching the selected genre; selecting 'All' resets the filter.

4.3 Score Tracking

- Each game session must submit a score to the backend when the session ends.
- The stored high score per user per game must update only when the new session score is strictly greater than the current stored value.
- The profile page must display the cumulative total score across all games played by the user.

4.4 Achievement Badges

- First Win: awarded when the user submits their first game score of any value.
- High Roller: awarded when the user achieves a score above 1000 in a single game session.
- Game Explorer: awarded when the user has played 3 or more distinct games.
- Favorite Collector: awarded when the user adds 5 or more games to their favorites list.
- Leaderboard Legend: awarded when the user holds a rank of 1, 2, or 3 on any per-game leaderboard.
- Each badge is awarded at most once per user and must not require admin action.

4.5 Leaderboards

- Per-game leaderboard: top 10 users by high score for that game, displayed as rank, username, avatar thumbnail, and score.
- Global leaderboard: top 10 users by cumulative total score across all games.
- Users with equal scores must receive equal ranks; the next rank must be skipped accordingly.

4.6 Admin Dashboard

- Add game: form accepts title, description, genre, thumbnail URL, and embed URL; saved game appears in the catalog immediately.
- Remove game: removes the game from the catalog; associated scores and leaderboard entries are retained in the database.
- Deactivate user: sets account status to inactive; deactivated users cannot log in.
- Admin dashboard is accessible only after admin login and must return HTTP 403 for unauthenticated requests to its API routes.

5. Non-Functional Requirements

Feature	Description
Performance	The homepage and game catalog must load within 3 seconds on a standard broadband connection. Game embeds load independently and are not included in this limit.
Security	All passwords hashed using bcrypt or equivalent. JWT tokens transmitted over HTTPS only. Admin routes return HTTP 403 for non-admin sessions.
Accessibility	All interactive elements must be keyboard-navigable. Images must include descriptive alt attributes. Color contrast ratio must meet WCAG 2.1 AA for body text.
Responsiveness	The site must render without horizontal scrolling or layout overflow at viewport widths of 320 px, 768 px, and 1280 px.
Browser Support	Chrome, Firefox, Edge, and Safari - latest two major versions of each.
Uptime	The deployed URL must remain publicly accessible for at least 14 days after delivery.

6. Screens / Modules

Module / Screen	Purpose
Homepage / Catalog	Displays all games in a card grid. Includes live search bar and genre filter.
Game Detail Page	Embeds the playable HTML5 game. Shows game title, description, user's current high score, and per-game leaderboard.
User Registration	Form for new account creation with email and password.
User Login	Authenticated login form with error messaging.
User Profile	Shows avatar, username, total games played, total score, and earned badges. Includes avatar upload and username change controls.
Favorites Page	Grid of games marked as favorites by the logged-in user with option to unfavorite.
Global Leaderboard	Ranked list of top 10 users by cumulative score across all games.
Admin Login	Separate login form for the admin account; accessible at a distinct URL path.
Admin Dashboard	Tools for adding and removing games, viewing the user list, and deactivating accounts.

7. Deliverables Summary

S.no.	Deliverable	Format
01	Complete source code	.zip archive
02	Deployed live URL	Publicly accessible link
03	README documentation	.md file
04	Environment variable template	.env.example file
05	Default admin credentials	Listed in README
06	Screenshot gallery	Minimum 5 PNG screenshots

8. Technical Notes

The following high-level technical constraints apply to this project. Implementation details are left to the developer.

- Frontend: React.js v18 or later, or Next.js v14 or later. Tailwind CSS or plain CSS/SCSS for styling.
- Backend: Node.js with Express.js or Django with Django REST Framework. A custom backend API is required; a frontend-only setup is not acceptable.
- Database: PostgreSQL or MongoDB. All data - users, games, scores, badges, favorites - must persist across sessions.
- Authentication: JWT-based. Tokens must be stored securely (HttpOnly cookies or equivalent). OAuth and social login are not required.
- HTML5 Games: Freely available open-source games or embeddable third-party games. No original game development is required.
- Hosting: Free-tier cloud hosting (Vercel, Render, Railway, Netlify, or equivalent) providing a public HTTPS URL.
- All configuration values (database connection strings, JWT secret, admin credentials) must be supplied via environment variables and must not be hard-coded in source files.